

Investigating the beneficial impact of segmentation-based modelling for credit scoring

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ARTICLE INFO

Keywords:

Credit risk modelling
Hybrid segmentation-based model
Logit leaf model
Interpretability

ABSTRACT

Due to its vital role in financial risk management, credit scoring has been investigated extensively in extant information systems studies. However, most credit scoring studies rely on one-size-fits-all classifiers with logistic regression (LR) as a popular benchmark. Moreover, extant literature largely focuses on predictive performance as an evaluation criterion. To find a better balance between predictive performance and interpretability though, the current study investigates the beneficial impact of segmentation-based modelling by benchmarking the logit leaf model (LLM) which is based on LR and decision trees. By a large experimental setup using a real-life credit scoring data set containing 65,536 active customers, we find that LLM is a viable classifier over its constituent parts, i.e., LR and decision trees, and is very competitive to state-of-the-art credit decision making techniques (neural networks, support vector machines, bagging, boosting and random forests) on three evaluation metrics (AUC, top-decile lift and profit). Furthermore, we show its extraordinary interpretability capacities by proposing a new visualization based on the LLM output. In sum, the excellence of the LLM as a classifier for credit decision making problems stems from its ability to combine strong predictive performance with interpretable insights that in turn can inform managerial decisions.

1. Introduction

Financial institutions rely on credit scoring systems to identify and minimise potential losses [7,16]. They thus can discriminate good from bad credit applicants by analysing available historical data, often with the assistance of statistical or machine learning techniques that support binary classifications [7,70]. Good applicants are those credit takers that repay their credit according to the terms of agreement; bad applicants instead default on their repayment obligations [30] and create losses that even may exceed the profits earned from good applicants [18]. Extant credit scoring literature proposes a wide range of predictive modelling techniques, which have been extensively developed and investigated in efforts to improve predictive accuracy.

Even a fraction of a percentage improvement in predicting customer credit default more accurately results in significant future profits [18,38,51]. Originally, it relied on statistical approaches such as logistic regression (LR) and discriminant analysis [4,25], then expanded to integrate machine learning techniques such as support vector machines (SVM) [39,41], k-nearest neighbours [38], genetic algorithms [62] and neural networks (NN) [21,77]. Recent contributions add complexity to

existing machine learning approaches, in attempts to improve classification performance, but in so doing, they seemingly ignore the need for interpretability [53,59,63,65], despite the substantial importance of interpretability for credit decision making.

In particular, explainability is a central and mandated requirement for credit decision making techniques [27]. For example, the standards of the Financial Stability Board [29] for global explanations of the credit scoring process mandate that they “demonstrate developmental evidence of theoretical construction, behavioural characteristics and key assumptions, types and use of input data, numerical analysis routines and specified mathematical calculations, and code writing language and protocols (to replicate the model)”. According to the “Ethics Guidelines for Trustworthy Artificial Intelligence”, each credit decision, even when based on complex data and algorithms, must offer transparency and instance-level explanations [26].

Extant literature has shown that hybrid, segmentation-based modelling approaches achieve both predictive performance and interpretability [19]. They connect and integrate different algorithms to obtain segment-based predictions, usually through a two-phase model. The first phase produces data subsets based on customers' similarities,

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and then the second phase entails training and predicting final risk probabilities with a classifier assigned to each subset. Unlike unsegmented black-box models, which necessitate post-hoc methods like SHAP [52], ICE [34] or LIME [68] to provide explanations after predictions have been made, the proposed techniques facilitate the generation of intrinsically interpretable, segment-centric predictions. Thus, interpretability is embedded during the model creation phase, aligning with regulatory prerequisites. Furthermore, this modelling approach is not just academically sound but has empirical backing. The relatively few studies that apply such modelling approaches confirm the greater scoring accuracy attained by segmenting the population, then predicting scorecards with a second classifier for each segment [39]. The resulting outputs are readily explainable on a segment level, because a classifier is trained on each segment. That is, hybrid, segmentation-based modelling promises output that is both accurate and interpretative, in accordance with the main requirements for credit scoring models. Yet recent credit scoring studies seemingly ignore these advanced, hybrid, segmentation-based modelling approaches [7,40].

The Logit Leaf Model (LLM), initially developed for customer churn prediction, segments customers using a decision tree and assigns them prediction scores on the basis of segment-specific LR. We leverage such research efforts to propose that LLM also can inform assessments of the creditworthiness of borrowers. In particular, we argue that LLM constitutes a hybrid, segmentation-based credit risk model, with proven merit in other predictive modelling contexts [17,19,20,45]. Therefore, we compare LLM's performance with nine benchmark models, using a real-life credit scoring data set of 65,536 active customers of a French consumer credit supplier. Given the importance of segment-specific insights for credit scoring, and to address recent calls for explainable analytics in OR, we also offer a new, clear visualization of LLM's hybrid output, showing how it can support decision-making within credit

scoring. Unlike the traditional LLM output visualization, the essence of the nomogram-based visualization is its ability to distill this complexity into a clear, intuitive, and user-friendly representation. By offering a visual interface that clearly demarcates segment-specific insights, this visualization technique bridges the gap between intricate model outputs and tangible, real-world decision-making. Moreover, in a domain where regulatory and ethical guidelines underscore the importance of transparency and explainability, the nomogram-based visualization serves a dual purpose. Not only does it enhance the interpretability of the model, but it also aligns with the mandates that emphasize the need for clear, replicable, and transparent decision-making processes.

Responding to the noticeable gaps in extant credit scoring literature, our study contributes in following ways. First, we endeavor to confirm the strong potential of hybrid segmentation-based models in harmonizing predictive accuracy and model's interpretability. We present the LLM as a competitive credit scoring algorithm based on its capacity to offer prediction performance without compromising on interpretability. Further highlighting the potential of the LLM, we propose a new nomogram-based visualization, adeptly transforming dense model outputs into transparent, intuitive, and easily interpretable visuals.

In Section 2, we offer a review of machine learning methods and interpretability techniques adopted in previous credit scoring studies. Section 3 details the proposed model, applied to a credit scoring context, and in Section 4, we describe the results of testing this model. On the basis of the detailed analysis in Section 5, we conclude in Section 6 with some implications and further research directions.

2. Literature review

Extant credit scoring literature contains a wide range of predictive modelling methods, as we summarise in Table 1. This review includes

Table 1
Literature survey (part 1/2).

Study	Interpretability		Modelling Approach		Model Base				Evaluation Metrics
	Reported ^a	Visualization	One-size-fits-all ^b	Hybrid segmentation ^c	DT	REG	SVM	NN	
Martens et al. 2010 [56]	X	X ^{d,e}	X		X		X		ACC
Twala 2010 [73]			X		X	X		X	excess error; error rate
Yu et al. 2010 [81]			X			X	X	X	ACC
Zhang et al. 2010 [82]			X		X		X	X	ACC
Zhou, Lai, Yu 2010 [85]			X		X	X	X	X	SE; SSP & PCC
Li et al. 2011 [48]			X		X	X	X		ACC
Finlay 2011 [30]			X		X	X		X	MR; Absolute & Relative improvement rates
Ping, Yongheng 2011 [65]			X			X	X	X	ACC
Wang et al. 2011 [76]			X		X	X	X	X	ACC
Yu et al. 2011 [80]			X		X	X	X	X	ACC
Akkoc 2012 [2]			X			X		X	ACC; AUC & MR
Li, Tsang, Chaudhari 2012 [49]			X		X	X	X		ACC; SE; SP & BA
Marques, García, Sanchez 2012 [54]			X		X	X	X	X	AUC
Marques, García, Sanchez 2012 [55]			X		X	X	X	X	AUC
Blanco et al. 2013 [10]	X	X ^d	X			X		X	AUC; MR
Kruppa et al. 2013 [43]			X		X	X			ACC; BS; SE & SP
Oreski, Oreski 2014 [63]			X					X	ACC
Verbraken et al. 2014 [75]			X			X		X	ACC; AUC; EMP and ROI
Lessmann et al. 2015 [47]			X		X	X	X	X	ACC; PCC; BS; H; PG & KS
Harris 2015 [39]			X	X		X	X		AUC
Zhao et al. 2015 [84]			X		X		X	X	Average & validation error
Lopez, Jeronimo 2015 [51]	X	X ^d	X		X				AR; AUC
Tomczak, Ziejaa 2015 [72]	X		X		X	X	X	X	TPR; TNR; Gmean; AUC

Notes: X = included.

^a Text explains drivers of bad creditors.

^b One classifier is applied to the whole training set.

^c One classifier is applied for segmentation, and another for the final classification.

^d Visualization by table.

^e Visualization by plot.

articles published between 2010 and 2021 from journals ranked in category 2 or higher in the Academic Journal Guide 2021 and focusing on experimental studies that demonstrate applications of credit scoring methodologies. The selected studies highlight four insights relevant to our research: whether the paper addresses the interpretability of the model decision (*Interpretability*), which type of modelling approaches it uses (*Modelling Approach*), the considered reference classifiers (*Model Base*) and the use of evaluation metrics (*Evaluation Metrics*). Each entry in this review explores the development, application and evaluation of credit scoring methods. For the interpretability dimension, we investigate whether the research effort zooms in on the model drivers and whether it provides a visualization to explain credit risk to managerial decision makers. The modelling approach column reflects whether the study used one-size-fits-all models or hybrid, segmentation-based models. For the model base, we identify whether the study uses one or more of the four most common credit scoring models (i.e., decision trees [DT], regression [LR], support vector machines [SVM], or neural networks [NN]). Lastly, the evaluation metrics list the metrics that are used to assess the performance of the models. (See Table 2.)

The *Interpretability* column lists studies that discuss the output of their proposed model, with in the *Reported* column, those studies that do report the outputs numerically (e.g. by means of regression coefficients), and those that depict it graphically are reported in the column *Visualization*. We find only a few studies that address the interpretability aspect, despite regulatory demands to explain models' output, as in the European General Data Protection Regulation and Basel agreements [14,28]. Only 23% of studies explicitly address the explainability of the scoring processes that their proposed methods use. Furthermore, less than 5% of studies (i.e. 20% of the 23% studies that include details on interpretability) include both textual and visual representations of their models' output (see Fig. 1).

The few studies that seek to report both on accuracy and

interpretability offer meaningful information about the decision logic involved in processing the data. For example, Fernandes and Artes [28] came to the conclusion that the risk of default relates to the economic environment, as gauged by the spatial dependence of defaulters. [57] use a Bayesian network with a latent variable and propose a calibration procedure. The proposed approach identified two segments of different probability of default. Those two segments share few similarities for both defaulters and non-defaulters. For instance, housing credit creates a higher probability of default compared with other credit types in both classes. And payment defaults appear more likely when the household members are under 35 years of age.

Next, we investigate the *Models* used; we do distinguish two types: *one-size-fits-all* models, such that the classifier is applied on the entire, global data set, or *hybrid, segmentation-based techniques* that first segment the data set, and then applies a classifier for each segment that provides scorecards for each individual applicant. As we noted previously, relatively few studies have adopted hybrid, segmentation-based techniques as which we list in Table 3. They offer high accuracy, because they segment the population on the basis of implicitly different risk profiles. For example, [39] proposes a clustered support vector machine model that, compared with nonlinear SVM, is computationally cheaper, even as it achieves similar prediction performance. He, Zhang, and Zhang [40] use a hybrid-sampling approach based on BalanceCascade. The original BalanceCascade approach was proposed by Liu, Wu, and Zhou [50] in order to obtain data with a similar number of positive and negative samples by randomly removing redundant samples with random undersampling techniques. The proposed Extended BalanceCascade approach (EBCA) is used to sample imbalanced data, then generate scorecards for each instance by using a different ensemble classifier for each subset. Each base classifier is optimised by the particle swarm optimisation algorithm, which improves the prediction result. The proposed model extends and outperforms the BalanceCascade approach

Table 2
Literature survey (part 2/2).

Study	Interpretability		Modelling Approach		Model Base				Evaluation Metrics
	Reported ^a	Visualization	One-size-fits-all ^b	Hybrid segmentation ^c	DT	REG	SVM	NN	
Yu, Yang, Tang 2016 [79]			X			X	X	X	ACC
Ala'raj, F. Abbod 2016 [3]			X		X		X	X	ACC; AUC; H-measure & BS
Leong 2016 [46]	X		X			X		X	ACC; SE & precision
Abelian, Castellano 2017 [1]			X		X	X	X	X	ACC & AUC
Beque, Lessmann 2017 [9]			X		X	X	X	X	PCC and AUC
Xia et al. 2018 [78]			X		X	X	X		ACC; AUC; AUC-H; BS
He, Zhang, Zhang 2018 [40]			X	X	X	X	X		AUC; Fmeasure; Hmeasure; G-mean; Logloss & KS
Kozodoi et al. 2019 [42]	X		X		X	X			EMP
Tobacka, Martens 2019 [71]			X				X	X	AUC; lift at 1% & 5%
Bao, Lianju, Yue 2019 [7]	X	X ^{d,e}	X	X	X	X	X	X	MCC; AUC; ACC; recall; precision
Papouskova, Hajek 2019 [64]			X		X	X	X	X	ACC; AUC; MC; R2; RMSE; MAE & profit
Masmoudi, Abid, Masmoudi 2019 [57]	X	X ^{d,e}	X		X		X		ACC; precision; recall & F1
Shen, Zhao, Kou 2020 [69]			X		X	X	X	X	AUC; KS & BS
Melo Junior et al. 2020 [59]			X		X	X	X	X	AUC; H; BAcc; G-mean; F1; F5; F35 & TPR
Djeundje et al. 2021 [23]			X		X	X		X	AUC
Gunnarsson et al. 2021 [35]			X		X	X		X	AUC; BS; PG & EMP
Benyacoub et al. 2021 [8]			X		X	X	X	X	accuracy; AUC
Bai, Zheng, Shen 2021 [6]			X		X			X	C-index; AUC; KS; Default rate
Medina-Olivares et al. 2021 [58]	X	X ^{d,e}	X			X			H-index; Gini; AUC; KS; Log sc
Zhang et al. 2021 [83]			X		X				ACC; BA; AUC; BS; F; Log sc
Ashofteh, Bravo 2021 [5]	X	X ^{d,e}	X		X	X	X	X	AUC; TPR; TNR; PPC; recall; & Log loss
study	X	X ^{d,e}	X	X	X	X	X	X	AUC; TDL & EMP

Notes: X = included.

^a Text explains drivers of bad creditors.

^b One classifier is applied to the whole training set.

^c One classifier is applied for segmentation, and another for the final classification.

^d Visualization by table.

^e Visualization by plot.

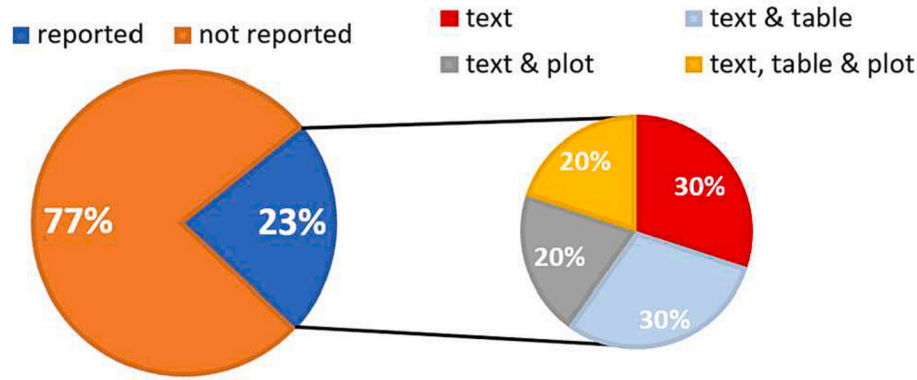


Fig. 1. Reported interpretability in credit scoring literature.

Table 3

Reported hybrid, segmentation-based approaches in credit scoring literature.

Study	Segmentation Method	Classifiers	Visualization
Harris 2015 [39]	K-means	LR/SVM-RBF/ SVM-linear	–
He, Zhang, Zhang 2018 [40]	Extended Balance	RF/XGBoost/	–
	Cascade Approach	LR/LDA/SVM/ KNN/DT	
Bao, Lianju, Yue 2019 [7]	Kmeans	LR/SVM/DT/ ANN/ KNN/RF/GBDT	Hierarchical clustering map
Our study	DT	LR	Decision rules

to generate adjustable, balanced segments based on the imbalance ratios obtained from the training set. [7] instead integrate supervised and unsupervised machine learning strategies at the consensus stage and data set clustering stage. In benchmarking their approach with various supervised algorithms, they confirm the superiority of the proposed integration. Despite these efforts, interpretability of the hybrid approaches remains a challenge, as most of these approaches rely on distance-based metrics to group customers rather than clear decision rules.

The *Model Base* column lists the base algorithms used in each study. 65.9% of the studies used at least three of the four most common credit scoring models. The authors either use a modified version of the model as their proposed approach, or they use the original version of the model as a benchmark approach, or both. For instance, Zhou, Lai, and Yu [85] proposed two groups of several SVM ensemble models. Those models aggregate least squares SVM (LSSVM) classifiers of individual members based on either the different weight assignment strategies, or the reliability of the decision made by each classifier. The proposed framework is tested on two real world datasets and benchmarked against 19 other techniques including 1-norm SVM with RBF kernel and with linear kernel. Zhang et al. [82] suggested a vertical bagging decision trees model (VBDTM) for assessing credit risks. The model is constructed as an improvement to the rough set and C4.5 method (RSC) by adding the strategy of vertical bagging. Their findings confirm that VBDTM is more robust and accurate than other types of model such as single models including SVM, NN, C4.5 and two stage model RSC among others. Kruppa et al. [43] applied an optimised logistic regression (logReg tuned) to a large dataset of complete payment data of short-termed credits. The proposed approach was outperformed by the Random Forest with probability estimation trees (RF-PET). Oreski and Oreski [63] evaluated the hybrid genetic algorithm with neural networks (HGA-NN) on a real-world credit dataset. The proposed approach is compared to 5 other techniques including NN and SVM, and it found to be a promising

alternative to identify an optimum set of both the neural network parameters and the feature subset for the credit risk assessment. Consequently, this study uses several techniques related to the four base algorithms listed in the *Model Base* as our benchmark techniques.

In terms of evaluation, most papers evaluate the performance of benchmark algorithms solely according to their predictive performance, without accounting for a profit-oriented business reality [36,37]. Only 4 of 44 studies we review suggest profit-driven model selection as a way to identify the most profitable model [35,42,64,75]. But such a selection approach can improve the practical application of the credit scoring models, by offering insights into possible monetary rewards or losses [75].

Considering the predominant focus on evaluating and improving predictive performance with one-size-fits-all models, credit scoring research that addresses interpretability is scarce. Noting the importance of interpretability at both global and segment levels, we seek to establish the value added by LLM, due to its capacity to achieve accuracy and profitability, as well as support clear visual depictions of the output.

3. The logit leaf model

The LLM was introduced to address churn by De Caigny, Coussement, and De Bock [19], as a hybrid, segmentation-based approach. It offers strong predictive performance, surpassing popular statistical and machine learning methods. It relies on a two-stage, hybrid classification algorithm. First, on the basis of similarities in individual characteristics, a DT generates n subsets of homogeneous groups. Second, the LLM applies n LR classifiers, depending on the number of generated subsets in the first stage, then a LR generates a scorecard for each and every member of the subset, as illustrated in Fig. 2.

From an implementation perspective, the LLM is attractive and offers multiple advantages [19]. First, the DT provides an overview of the characteristics of the generated segments. Second, the scorecards in the final phase are easy to explain. This combination establishes the interpretability required by legislative and operational authorities.

The merits of LLM also have been established in various studies [17,19,20,45], in terms of both power and interpretability. [45] conduct a comparative study of two customer churn data sets, such that they compare the performance of LLM with a new hybrid algorithm called the support leaf model (SLM). According to the area under the receiver operating characteristics curve (AUC) and top-decile lift (TDL), LLM outperforms SLM. [17] benchmark LLM against eight different algorithms, using a real-life data set of online students and attempting to predict their dropout probability. They confirm that LLM offers the best trade-off between prediction performance and interpretability and further note that the LLM's visual depictions provide quick, deep insights into student dropout behaviours on both segment and student levels. [20] introduce an uplift version of the LLM for uplift prediction in a business-to-business context. Their findings affirm the superiority of its

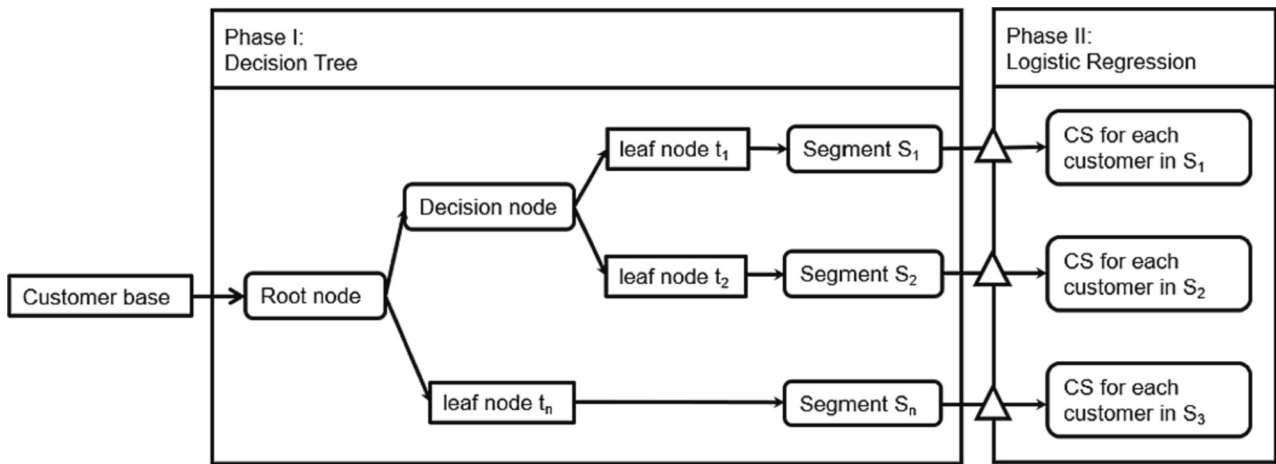


Fig. 2. Overview of the LLM algorithm.

performance, as well as the viability of the insights provided by this proposed method. Credit decision processes also require explainability to satisfy regulatory requirements, so we propose LLM as a potentially significant decision support tool that can satisfy financial regulators' needs for interpretability while also providing evidence to support managerial decisions and ensure good performance.

4. Experimental design

4.1. Data

We conduct a real-life experiment, using data from a European financial institution that offers a variety of services, including customer credit. The data set contains 65,532 customers and 90 features that describe the credit characteristics and payment behaviour of each customer. Defaulters account for 48.29% of the total number of customers; they failed to repay at least one of the credits they received, according to their borrowing agreements. This category includes all unjustified nonpayments, starting from the due date, such as late payments but also rearranged payments. With a binary measure, we assign a value of 0 if the customer succeeded in paying back the credit and 1 if they defaulted. The features are observed over a six-month window after the date of the last granted loan, so the date varies across customers. If a customer does not pay back the credit in the following six months, the company defines him or her as a bad creditor.

Table 4 summarizes the 90 independent features considered in this study, as well as an exact count and examples for each category. These independent features span a two-year observation period, such that we can capture clients' characteristics and categorise them according to payments made, purchasing activities, credit details, monetary report and consumed services. Each category consists of multiple features; the category with the most features is *Monetary*, which represents the amount of money spent by type of product.

Table 4
Independent features.

Feature Category	Examples	# of Features
Payments	Number, dates and duration of unpayments	20
	Length of relationship, segmentation of the	21
Customer activity	population by activity/period, revoke, type of product and management	18
	Usage and evolution of usage of the credit reserve	
Credit	Amount of money spent by type of product	24
Monetary	Number of services used by period	7
Services		

4.2. Experimental setup

We performed preprocessing of the data set, in line with extant credit scoring literature. First, we apply established missing value replacement guidelines [47] and treat missing values differently, depending on the variable type. Continuous variables are imputed by the mean; missing values of categorical variables instead are replaced by the mode. All categorical variables also are transformed into $c - 1$ dummy variables. After the data preprocessing, we filter out redundant variables from the data set using the Fisher score. A subset of 20 variables with the highest Fisher scores, indicating the best predictive power, then undergoes further processing [74].

After preprocessing the original data, we implement a 5×2 cross-validation (CV). In each iteration, the data set is randomly and equally split into two folds. We build the models using 2/3 of each fold as the training set, then validate performance using the remaining 1/3 of the same fold as a hold-out. Then we evaluate the best combination of hyperparameters of each fold using the other fold as test set [22]. Following on our literature review, we include ten benchmark algorithms: Logit Leaf Model (LLM) [19]; Logistic Regression (LR) [24]; Decision Tree C4.5 (C4.5) [66]; Classification and Regression Trees (CART) [13]; Boosted Decision Trees (BoostedTree) including Adaptive, Logit and Gentle Boosting Trees [31,32]; Bagged Decision Trees (BaggedTree) [11]; Random Forest (RF) [12]; Feed-forward Multilayer Perceptron Neural Network (NN) [67]; Support Vector Machines (SVM) [60]; Logistic Model Tree (LMT) [44]. The selected methods have been considered in previous studies and represent the state-of-the-art techniques for credit scoring [47]. The hyperparameter value ranges, as listed in Table 5, are based on previous, large-scale benchmarking studies [7,42,47,59,69,78].

According to our literature review, predictive performance is a key requirement for credit scoring models. Statistical evaluation metrics are frequently used, as opposed to more managerial evaluation metrics such as TDL or expected maximum profit (EMP) [35,42,71,72,75]. Therefore, we include two non-statistical evaluation dimensions to assess the credit scoring models. First, we gauge its discriminatory ability to establish the likelihood of defaulting, on an individual and a proportional level, using the AUC [47] and TDL [61]. The AUC represents the probability that a randomly selected defaulter is ranked higher than a randomly selected non-defaulter by a classifier. It provides an aggregate measure of a model's performance across all classification thresholds. It is computed using following formula:

$$AUC = \int_0^1 \frac{TP}{AP} d \frac{FP}{AN} \quad (1)$$

Table 5
Hyperparameter settings of considered models.

Algorithm	Parameter	Meta-Parameter	Candidate Settings
LLM	threshold_pruning	Confidence threshold for pruning	0.01, 0.1, 0.15, 0.2, 0.25, 0.3
	nbr_obs_leaf	Min. no of observations per leaf	n*(0.01, 0.025, 0.05, 0.1, 0.25, 0.5)
LR C4.5	direction	Direction	forward, backward, both
	C	Confidence threshold for pruning	0.01, 0.1, 0.15, 0.2, 0.25, 0.3
	M	Min. leaf size	n*(0.01, 0.025, 0.05, 0.1, 0.25, 0.5)
CART	minbucket	Min. leaf size	n*(0.01, 0.025, 0.05, 0.1, 0.25, 0.5)
BaggedTree	prune	Pruning	on/off
	bw.iters	No. of fitted models in bagging	10, 20, 30, 40, 50, 100, 250, 500, 1000
BoostedTree	algorithm	Algorithm	AdaboostM1, Logitboost, Gentleboost
	iter	No. of boosting iteration	10, 50, 100, 250, 500, 1000
	shrinkage	Learning rate (Gentleboost)	0.1, 0.5, 1
RF	ntree	No. of CART trees	100, 250, 500, 750, 1000
	mtry	No. of randomly sampled variables	$\sqrt{m(0.1, 0.25, 0.5, 1, 2, 4)}$
SVM	cost	Regularization penalty	$2^{(-12, -13, \dots, 12)}$
	gamma	Width of RBF kernel	$2^{(-12, -13, \dots, -1)}$
NN	hidden	No. of hidden nodes	2, 3, ..., 20
	l1	Regularization penalty	$10^{(-4, -3.5, \dots, 0)}$
LMT	–	–	–

where TP (True Positive) denotes loans accurately categorized defaulting; FP (False Positive) signifies the instances where defaulters are incorrectly classified as non-defaulting; AP represents the actual count of defaulting loans; AN stands for the actual number of loans that did not default.

Further, the TDL measures how much better a prediction model is at identifying defaulters compared to a random model. Specifically, it calculates the *lift* for the top 10% of observations predicted to default. The TDL is defined using the TP and FP metrics of the confusion matrix calculated using the top 10% of observations predicted to default as:

$$TDL = \frac{TP_{top10\%} / (TP_{top10\%} + FP_{top10\%})}{AP / (AP + AN)} \quad (2)$$

Lastly, we measure the most profitable classifier using the EMP criterion, which is expressed as a percentage of the joint probability density of the classification costs [75]. This metric allows to connect the return on investment (ROI), which considers both costs and revenues, the prior probabilities of a loan being repaid (π_0) and not being repaid according to the repayment obligations (π_1), and the classifier's ability to rank prospects (AUC). The metric is identified as following:

$$EMP = 2 \cdot ROI \cdot \pi_0 \pi_1 \left\{ AUC - \frac{1}{2} \right\} \quad (3)$$

5. Results & discussion

To start, we present the predictive modelling results of the different algorithms and compare them with the LLM, according to three performance measures (AUC, TDL and EMP). Then we outline the

visualization of the LLM from Table 8 and compare it with the LR output in Table 7. Finally, we introduce a new visualization, as illustrated in Fig. 3, that offers deeper insights into default behavioural drivers.

5.1. Predictive performance

Table 6 contains the cross-validated results and details performance in terms of AUC, TDL and EMP. The values represent cross-validated average performance; the standard deviations are in brackets. We compare the predictive performance of LLM against nine benchmark models, using the real-life credit risk data set. The benchmark models can be classified into four categories: building blocks of LLM (LR and C4.5), four tree-based methods (BaggedTree, BoostedTree, CART, and RF), two advanced methods that often are used in credit scoring (NN and SVM), and a similar two-step method (LMT). In the table, the best performing method is underlined, for each evaluation metric. To determine the significant differences between LLM and the benchmark models, we use the post-hoc test based on Friedman's Aligned Ranks test [33].

The results affirm the strong performance of LLM on all three evaluation metrics. It achieves significantly better performance than BaggedTree, CART and C4.5. In addition, LLM is competitive with state-of-the-art ensemble methods like BoostedTree and RF, in addition to the LMT. It never performs significantly worse than the best performing models, and it has even the best performance on the managerial relevant TDL in absolute numbers. These cross-validated results confirm that LLM is a valuable classifier for the credit scoring domain.

5.2. Interpretability

The LR represents one of the building blocks of the LLM, as well as the golden standard of the credit scoring industry due to its simplicity and intrinsic interpretability [15]. Table 7 contains the visualization of the LR. It includes the selected features. The unique segment in this case represents the full data set, with a default rate of 48.52%. The results indicate that the variables with the most influence are V3, V11 and V15. Those variables represent information about granted split payments loans, nominal rate, and renewable loans, respectively. Both granted split payments loans, nominal rate have a positive influence, meaning that those features decrease default probability. Whereas renewable loans may increase default probability. The findings are in line with the business expectations, since both granted split payments loans and nominal rate help to update the customer's profile and adjust his payment capacities. While renewable loans are different than traditional loans, so it can be confusing to customers who are unfamiliar with this type of loan.

By showcasing the interpretability of LLM on global and local levels, we seek to offer deeper insights to financial decision makers. We present both an original LLM visualization and a new option, to expand these managerial insights. The former representation, as presented in Table 8, projects the original LLM output from the first fold of the experimental design (see Section 4). The results then get compared to the output of the LR from Table 7, reflecting the status of LR as a conventional credit scoring approach for estimating risk scorecards ([7,18]). It offers transparency and interpretability, to meet the requirements of financial regulators [7]. The new LLM visualization extends the original version by providing deeper insights into the specific segments, which is relevant for the credit decision making process.

In Table 8, we represent the results of the first phase of the LLM (DT). The table includes the decision rules that generate the segments, together with the number of observations and incidence rate for each segment. Then the results of the second phase are represented in Table 9. The table indicates the intercept as well as the retained features for each segment and their LR feature coefficients. These features are grouped according to whether they are shared over the different segments or unique for a specific segment. The shared, retained features over the different segments support comparisons across segments; the unique

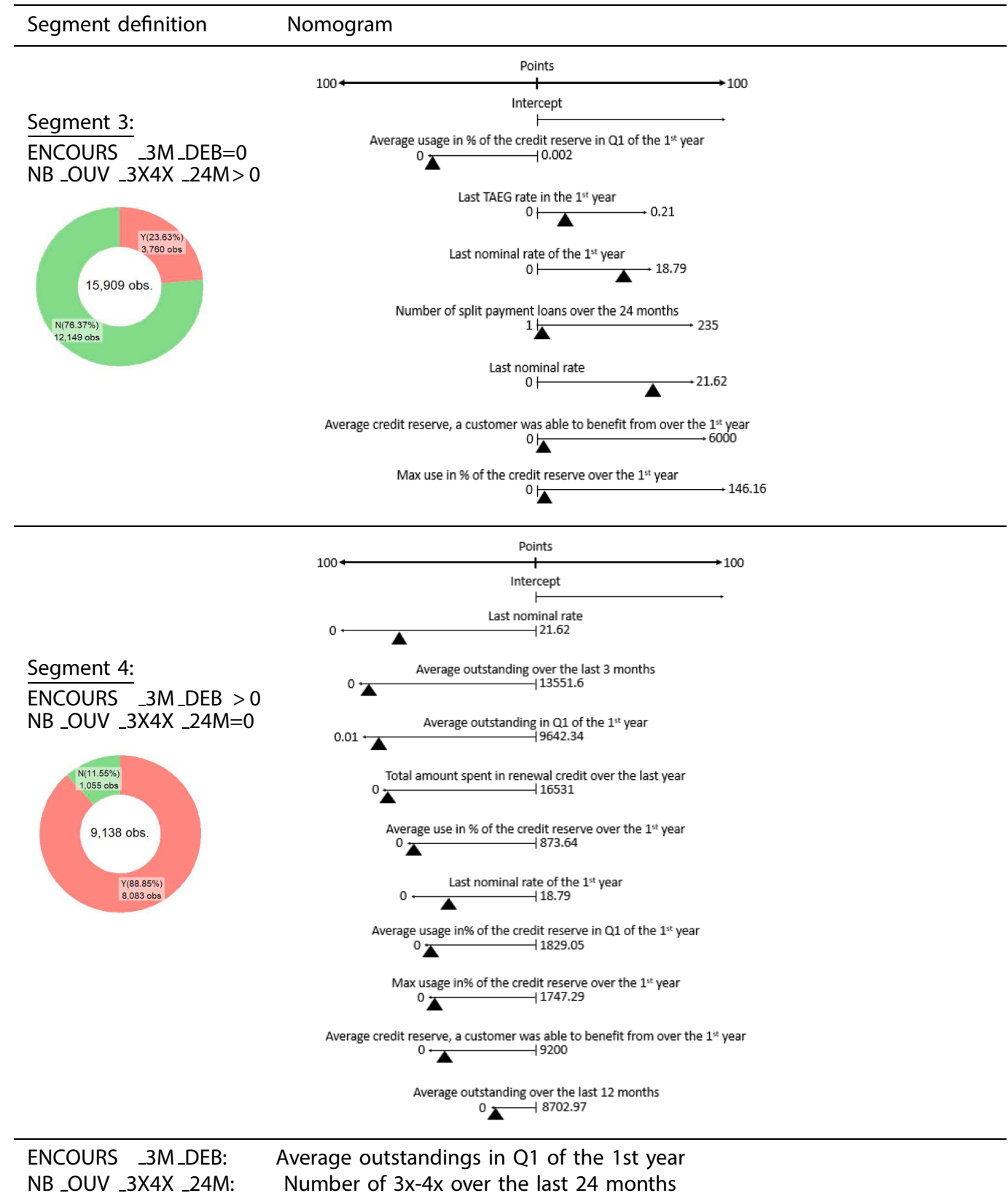


Fig. 3. LLM: New visualization.

segment-level features offer more profound insights for managerial decisions.

As Table 8 shows, the LLM identifies six segments based on three features. Segment 1 includes 26.95% of bad creditors, compared with

64% defaulters in Segment 2. The customers in both these segments have no outstanding credit in Q1 of the first year ($V1 = 0$). However, customers in Segment 2 had a credit reserve to benefit from over the first year ($V2 > 0$). The type of credit they received is renewable, so

Table 6

Predictive performance overview.

Model	AUC(SD)	TDL(SD)	EMP(SD)
LLM	0.816(0.002)	1.962(0.0007)	0.08930(0.01)
LR	0.792(0.001)***	1.935(0.0007)	0.08764(0.00)***
C4.5	0.799(0.004)***	1.660(0.0007)***	0.08827(0.19)***
CART	0.796(0.014)***	1.595(0.0008)***	0.08837(0.33)***
BaggedTree	0.800(0.003)***	1.750(0.0007)***	0.08875(0.03)***
BoostedTree	0.816(0.001)	1.924(0.0006)	0.08932(0.04)
RF	0.817(0.001)	1.956(0.0005)	0.08929(0.01)
SVM	0.804(0.002)***	1.917(0.0005)	0.08905(0.00)
NN	0.811(0.005)	1.849(0.0005)*	0.08911(0.18)
LMT	0.816(0.001)	1.961(0.0007)	0.08947(0.18)

***, **, and *, indicate significance differences with LLM at the 99%, 95%, and 90% confidence level, respectively.

outstanding credits get updated each time the customer uses more money. Segment 2 customers appear not very familiar with this type of credit and exhibit an increasing likelihood of defaulting. The customers in Segments 3 and 4 are distinguishable on two features: average outstanding amount over Q1 of the first year (V1) and the number of split payment credits over the complete period (V3). We observe default rates of 23.63% in Segment 3 but a defaulting rate of 88.45% in Segment 4. Although the customers of Segment 3 have no outstanding credit in Q1 ($V1 = 0$), they exhibit a positive number of split payment credits over the entire period ($V3 > 0$). In contrast, clients in Segment 4 might have an outstanding amount ($V1 > 0$), but they were never granted split payment credits over the course of two years ($V3 = 0$). The decision rules of the former segment selects customers that may be better profiled, due to the granted split payment applications. The final two segments can be partitioned on the basis of the same two features (V1 & V3), but customers in both these segments show a positive outstanding amount in Q1 ($V1 > 0$) and different ranges of granted split payment credits (V3). For example, customers in Segment 6 appear to engage in better management of the reimbursement of their credits, which lowers their default rate. The customers in Segment 5 instead seem to struggle to manage several credits simultaneously ($(V1 > 0) + (0 < V3 \leq 2)$).

To support decision-making, the second phase of the LLM is represented in Table 9, and it clarifies the differences across segments on the basis of various levels: a comparison between segments using the LR beta coefficient of the shared retained variables of the LR classifiers (*Shared Variables*); and an identification of segment-specific drivers of defaulting (*Specific variables*). We uncover some interesting insights by comparing the segments with the lowest and highest default rates (Segments 3 and 4, respectively).

We can also compare the coefficients of the shared features of the same two segments. Some shared features have similar impacts on the default probability in both segments, whereas others have inverse effects. For example, we observe a positive coefficient for feature V10 in Segment 4, but a negative coefficient of the same feature for Segment 3. The feature represents the last nominal interest rate of the first year. This finding confirms that customers of this segment are better profiled due

to the granted split payment credits ($V3 > 0$), which allows the profile to be updated. Consequently, the nominal interest rate can be set according to the updated profiling of each customer, which should reduce the default probability. Then we observe an inverse effect of feature V11, which represents the last nominal rate of the entire period, for these two segments. The evolution of Segment 4 follows recovery phases, which can include adaptations to the characteristics of the credit, such as the nominal rate. But the change in Segment 3 is likely due to the lack of an updated nominal interest rate, even as more credit is granted. Therefore, updating the nominal interest rate each time a customer increases their debt can reduce the default probability for this segment. Moreover, we observe a negative coefficient for feature V13 and a positive coefficient for feature V15, which represent the relative maximum usage of the credit reserve in Q1 and in the entire first year of the period, respectively. The first coefficient affirms that a convenient amount of money was granted as credit reserve, due to an accurate profiling of customers. Thus reconfirming our previous findings. However, this amount limit does not get updated according to the customer's total debt, which can increase during the year. In Segment 4, we note a positive coefficient for both variables that represent the granted credit reserve, which again confirms the results of our previous deductions about not profiling the customer or updating the credit reserve limits, even as the outstanding amount evolves with additional granted credit applications.

Actionable insights also can be derived from the segment-specific retained variables. For the segment with the highest default rate (Segment 4), we identify a positive coefficient for V8 and a negative coefficient for both V1 and V9. These variables represent the average outstanding amount over Q1 of the first year, the second year, and the last three months, respectively. Those outstanding amounts represent either accumulated loans over two years or long-term loans. In both cases, customers of this segment tend to repay their loans in the first year, but they start to default around the second year, with an exception in the last three months. The exception is due to recovery plans suggested by the firm. Therefore, earlier suggestions of recovery options to customers in Segment 4 might decrease their default rate.

The new visualization of the LLM presents a brief summary of key

Table 8

Predictive performance overview of the LLM: First step: Decision Tree.

First step: Decision Tree					
Seg	Rule1	Rule2	Rule3	# obs.	Incident
1	V1 = 0	V2 = 0	V3 = 0	1302	26.95%
2	V1 = 0	V2 > 0	V3 = 0	2020	64.05%
3	V1 = 0		V3 > 0	15,909	23.63%
4	V1 > 0		V3 = 0	9138	88.45%
5	V1 > 0		0 < V3 ≤ 2	2763	63.69%
6	V1 > 0		V3 > 2	1636	39.97%
V1	Average outstandings in Q1 of the 1st year				
V2	Average credit reserve amount, a customer was able to benefit from over the 1st year				
V3	Number of split payment loans over the last 24 months				

Table 7

Predictive performance overview of the LR.

Seg	# obs.	Incident	Incpt	V1	V2	V3	V4	V9	V11	V12	V13	V15	V16
1	32,768	48.52%	-0.11	-0.00	0.00	-0.01	-0.00	-0.00	-0.04	0.00	0.00	0.01	-0.00
V1	Average outstandings in Q1 of the 1st year												
V2	Average credit reserve, a customer was able to benefit from over the 1st year												
V3	Number of 3×-4× over the last 24 months												
V4	Total amount spent in renewal credit in last year												
V9	Average outstanding over the last 3 months												
V11	Last nominal rate												
V12	Average usage in % of the credit reserve in Q1 of the first year												
V13	Max usage in % of the credit reserve in Q1 of the first year												
V15	Max usage in % of the credit reserve over the first year												
V16	Average amount spent in 3×-4× over the last 24 months												

Table 9
Predictive performance overview of the LLM: Second step: Logistic Regression.

Seg	# obs.	INDT	Second step: Logistic Regression																	Specific variables																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
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characteristics on the segment level, as is illustrated in Fig. 3. Analysing drivers on the segment level is important for creating segment-specific targeting strategies. It differs from existing visualisations in three respects. First, the new visualization depicts a definition of each segment. Second, it provides a chart that enumerates bad creditors in each segment. Third, it offer a nomogram of the retained features of the respective segments. Establishing this new visualization involves the following steps:

- Standardise all features on a similar scale and represent their relative impact on the predicted likelihood of defaulting.
- Features with a negative beta coefficient take an arrow pointing left (i.e., higher value increases the predicted defaulting behaviour), and features with a positive beta coefficient have an arrow pointing right (i.e., higher value decreases predicted defaulting behaviour).
- Sort features from most to least impact on segment defaulting behaviour, such that features that increase the likelihood of defaulting appear first. The impact is represented by the length of the arrow. The original range of the feature (minimum and maximum values) is indicated by each end of the arrow.
- Use a pointer (▲) to indicate the feature average value for the segment.

This visualization offers deeper insights into the relevance of different drivers of default. We note that Segment 4 groups most of the bad creditors and shows negative beta coefficients for all its impactful features. Segment 3 instead groups most of the good creditors, with positive coefficients for almost all its retained features. The more the pointer deviates from the middle, the greater the feature's influence on customers' predicted behaviours. For example, the model predicts that average usage, as a percentage of the credit reserve in Q1 of the first year, is the main driver of default in Segment 3. An intervention is needed to get members of this segment back on track, such as offering them techniques for using their granted credit reserve without falling into nonpayments or encouraging them to make regular payments before using more of their reserve. Moreover, unlike the LR, the LLM can identify the different segments of customers who can be affected (positively or negatively) by the retained features by the LR (Table 7), as well as other ones. Thus, we show that the LLM can help decision makers identify and tackle drivers of defaulting behaviours at a more granular, segment level, instead of for the entire data set, as occurs with LR. In turn, the LLM offers deeper insights into the drivers and supports for more fine-grained decisions about which actions to take to reduce credit risk. Our findings identify LLM as the algorithm that best balances classification performance with interpretability.

To deliver additional insights into the functioning of the LLM model, we conducted two sensitivity analyses building on the best hyper-parameter test set results of the 5×2 cross-validation. Fig. 4 examines the interactions between the LLM's overall prediction performance on the Y-axis (*AUC*), the number of DT segments of the LLM (*Nb segs*), along with LR variable characteristics of the LLM represented on the X-axis. Each data point in the panels of Fig. 4 represents one fold of the 5×2 cross-validation. Panel a represents on the X-axis the average number of variables used in the LR models in the LLM (*Average number of variables*), while panel b shows the average number of variables across the LR models in the LLM common to the set of variables of the best performing LR model (*Average number of common variables*). Several conclusions emerge from these sensitivity analyses. First, LLM models with an increased number of segments tend to perform better as seen in both panels. In other words, if the DT succeeds in finding more fine-grained segments in the first phase of the LLM, it tends to benefit the overall LLM prediction performance. Second, there is a tendency that across segment sizes having more variables retained in the LR models of the LLM, the performance of the LLM is better. Third, we observe that the overlap of common variables in the LR models in the LLM with the best performing LR does not have an impact on the prediction performance.

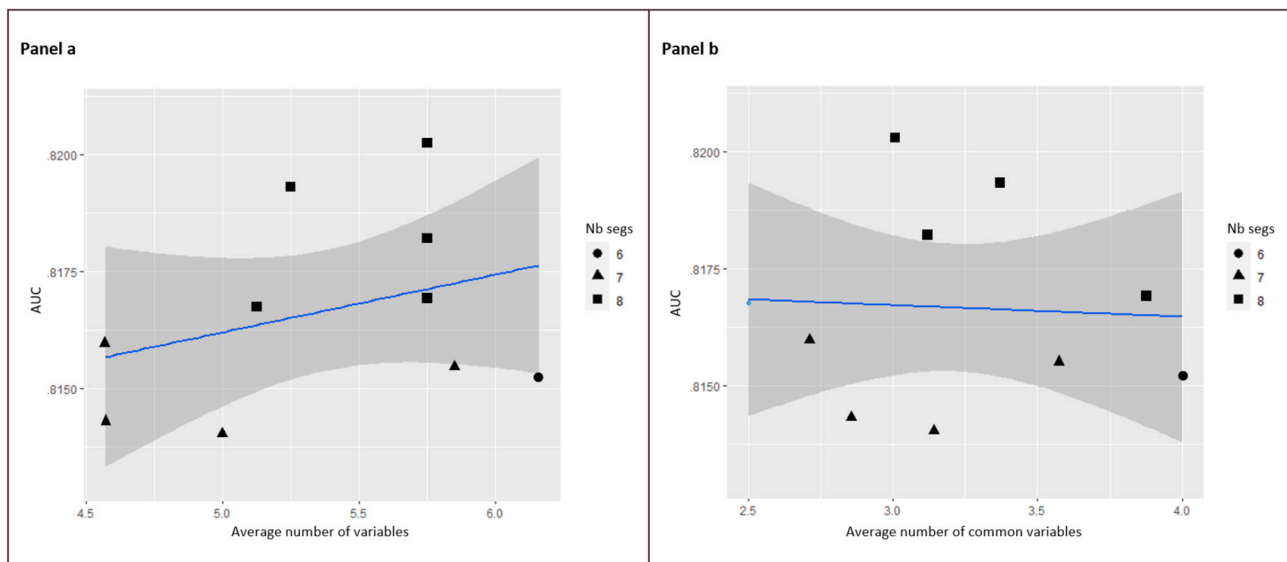


Fig. 4. Sensitivity Analyses.

These conclusions suggest that the LLM model's performance is optimised when the LLM succeeds in delivering a more fine-grained segmentation with more segments and when more variables are retained in the second phase of the LLM.

6. Conclusion and further research

One of the prime obligations of banking supervisors in many countries is to justify decisions on credit applications [18,51,72]. Therefore, a credit decision making model's comprehensibility is as important as its accuracy in terms of discriminating good from bad creditors. To address both demands, we introduce the LLM as a hybrid, segmentation-based approach that achieves a good balance between classification performance and interpretability. Using a real-life data set of 65,536 customers and 90 features obtained from a French consumer credit supplier, we benchmark LLM against state-of-the-art credit scoring models.

This study thus contributes to extant literature and practice by offering decision-makers more in-depth insights into the drivers of customers' default, as well as solutions to help them manage credit risks. First, the benchmarking results confirm that LLM offers an excellent trade-off between prediction performance and interpretability. It outperforms its building blocks, LR and C4.5, and is competitive with popular tree-based ensembles and more advanced methods like NN and SVM. Second, LLM provides deeper insights into the prime drivers of default on a customer segment level. Although hybrid segmentation modelling has drawn little consideration in credit scoring literature thus far, our study confirms its strong potential for balancing predictive performance with output explainability. Third, we propose two visualization options that satisfy financial legislative requirements, as well as help decision-makers design relevant default prevention actions that diminish their credit risks.

Although this study thus contributes to credit risk decision support systems literature, we also note several paths for further research. First, we used a real-world, unique data set to introduce the LLM to the credit scoring field. Despite the advantages of real-world data, they remain difficult to acquire, so we were constrained to studying only a single context. Additional research is needed to explore the beneficial effects of LLM in other credit scoring contexts. Second, we propose an extended visualization that is based on nomograms and that can profile customers, according to their predicted credit risks. We present clear credit analysis for decision makers, but more research is needed to address cases in which the number of retained features becomes too large, making the

tables less interpretable. A valuable path for research would be to find creative ways to present the outputs of LLM when output dimensionality is substantial.

CRediT authorship contribution statement

Khaoula Idbenjra: Conceptualization, Data curation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. **Kristof Coussement:** Conceptualization, Methodology, Supervision, Visualization, Writing – review & editing. **Arno De Caigny:** Conceptualization, Methodology, Supervision, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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